

Project Overview

EEG research typically relies on rigid commercial tools or fragile ad-hoc scripts, making it difficult to reproduce results and share pipelines across teams.

This project delivers a **full-stack EEG analysis platform** that standardises every step from raw data to model output — built for domain researchers, not just engineers.

Objectives

- Build a systematic, reproducible EEG analysis platform usable by domain researchers without deep engineering expertise.
- Evaluate whether **2-second prestimulus EEG** can predict:
 - ① Trial correctness (classification)
 - ② Reaction time (regression)
- Provide transparent experiment tracking so results can be replicated and extended.

Platform & Visualization

The platform exposes a modal workflow — users select a *mode* (e.g. Preprocess, Epoch, Analysis) then configure and execute actions via a clean web interface. All outputs are cached and parameterised for full reproducibility.

```
<div class="ui-body">

  <!-- Left: main UI screenshots -->
  <div class="ui-main">
    
    
  </div>

  <!-- Right: 2x2 plot grid -->
  <div class="ui-plots">
    <figure>
      
      <figcaption>Evoked Topography</figcaption>
    </figure>
    <figure>
      
      <figcaption>Frequency Spectrum</figcaption>
    </figure>
    <figure>
      
      <figcaption>Power Spectral Density</figcaption>
    </figure>
    <figure>
      
      <figcaption>Epoch Overview</figcaption>
    </figure>
  </div>

</div>
```

Research Findings

Study Setup

- Dataset:** Healthy Brain Network EEG (HBN-EEG, BIDS, 128 ch)
- Task:** Contrast Change Detection (CCD) — 2-second prestimulus window
- Model:** EEGNet (end-to-end deep learning)

Predictive Models

- Classification:** Correctness prediction showed limited separability; class imbalance mitigation improved recall but not overall accuracy.
- Regression:** Reaction time prediction showed weak fit, suggesting 2-second prestimulus window is insufficient alone.
- Insight:** EEG prestimulus signals contain some behavioural information, but stronger models and richer features are needed.

Future Work

- Model improvements:** Transfer learning and subject-adaptive fine-tuning to overcome class imbalance and high inter-subject variability.
- Feature engineering:** Time-frequency representations (ERSP), functional connectivity, and source localisation as additional inputs.
- Platform extensions:** Real-time streaming support, multi-task learning modules, and automated hyperparameter search.
- Dataset expansion:** Broaden beyond CCD to other HBN tasks to test generalisability of the platform and models.